Database Systems(CSL3050)

Lab Assignment 10 - Report

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1 Introduction

This report presents Cypher queries written for the Neo4j database related to the Movie Recommendations dataset. The assignment involves answering specific questions by querying the graph database.

2 Query 1: Actors in Selected Movies

The first task is to write a Cypher query to return all actors who played a role in any of the movies: "Inception", "Interstellar", or "Dunkirk".

```
MATCH (a:Actor)-[:ACTED_IN]->(m:Movie)
WHERE m.title IN ["Inception", "Interstellar", "Dunkirk"]
RETURN a.name AS actor, m.title AS Movie
```

Output:

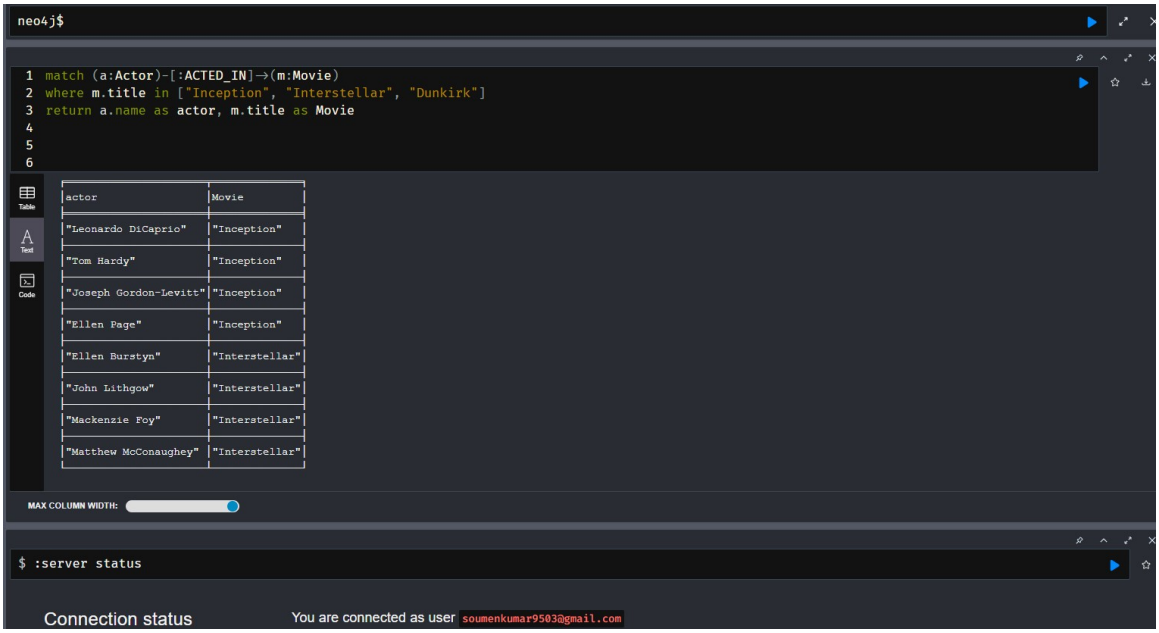


Figure 1: Output for Query 1: Actors in Selected Movies

3 Query 2: Add Director and Composer

In this task, the goal is to add a new Director and Composer for the movie "Forrest Gump" . If the Director and Composer nodes do not exist, create new nodes for them.

```
// Adding the director
MERGE (d:Person {name: "Robert Zemeckis"})
SET d:Director
WITH d
MATCH (m:Movie {title: "Forrest Gump"})
MERGE (d)-[:DIRECTED]->(m)

// Adding composer and creating relationship
MERGE (c:Person {name: "Alan Silvestri"})
SET c:Composer
WITH c
MATCH (m:Movie {title: "Forrest Gump"})
MERGE (c)-[:COMPOSED]->(m)
```

Output:

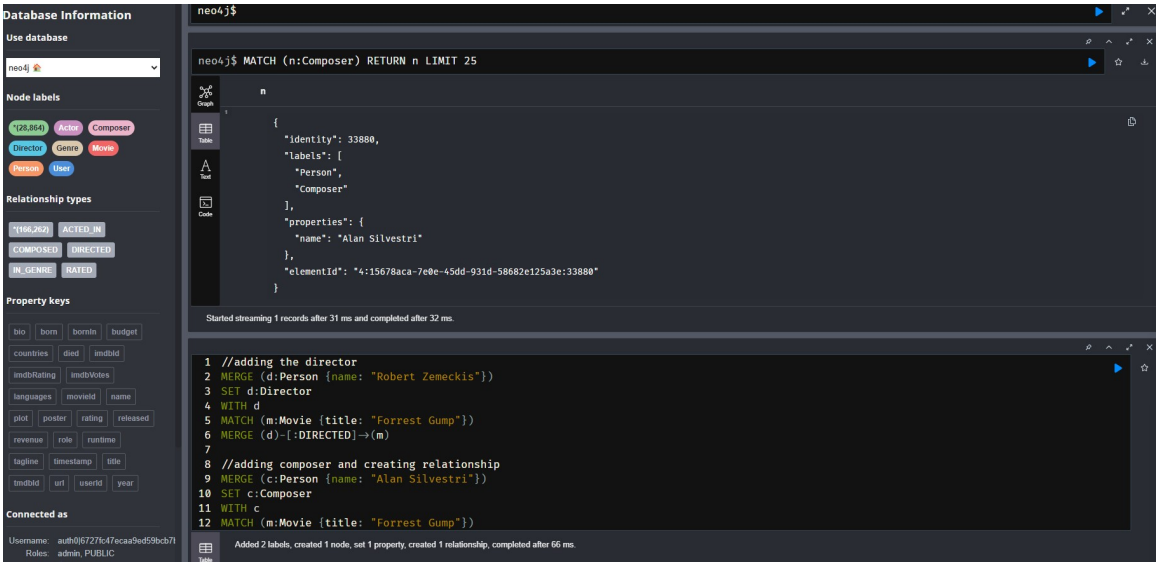


Figure 2: Output for Query 2: Add Director and Composer

4 Query 3: Movies Based on Rating

This query returns movies that have a rating of more than 8 or are unrated (do not have a rating relationship).

```
MATCH (m:Movie)
WHERE m.imdbRating > 8 OR m.imdbRating IS NULL
RETURN m.title AS MovieTitle, m.imdbRating AS IMDbRating
```

Output:

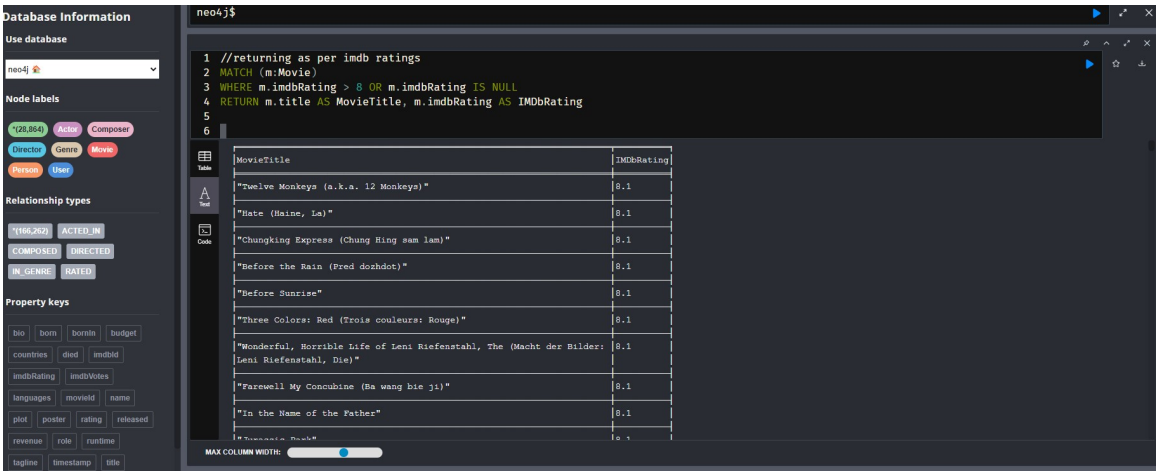


Figure 3: Output for Query 3: Movies Based on Rating

5 Query 4: People in Multiple Roles

This query returns all people who have been both an Actor and a Writer for at least one movie (not necessarily for the same movie).

```
MATCH (p:Person)-[:ACTED_IN]->(:Movie)
MATCH (p)-[:DIRECTED|:COMPOSED]->(:Movie)
RETURN DISTINCT p.name AS PersonName
```

Output:

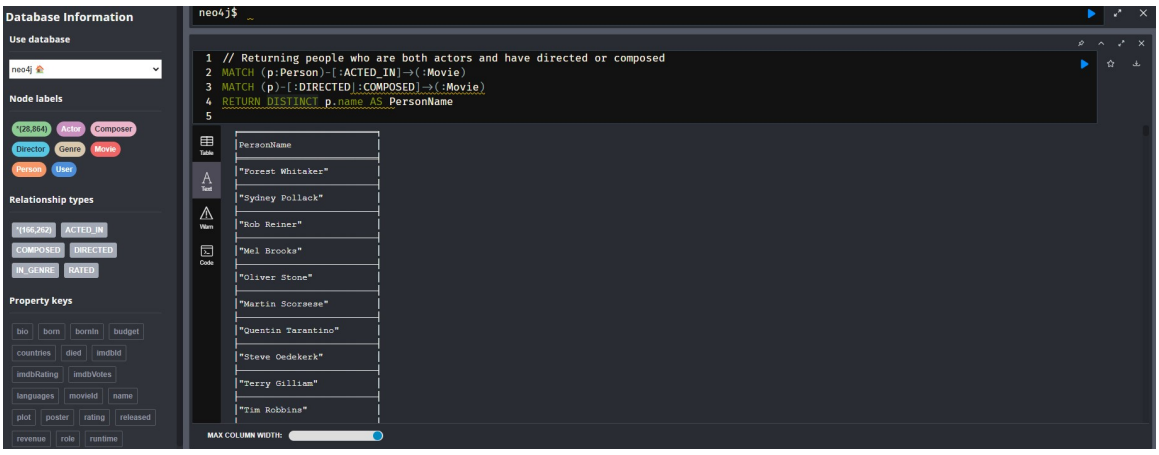


Figure 4: Output for Query 4: People in Multiple Roles

6 Conclusion

The queries presented above demonstrate how to interact with a graph database using Cypher to extract useful insights about movies, actors, and related entities such as directors, composers, and ratings. The results obtained are essential for understanding relationships in a movie recommendation system.